

Naive Bayes Classification Report

Dataset: Mushroom Dataset

Train–Test Split: 80/20

Part 2A Evaluation of Custom Naive Bayes Implementation

The Naive Bayes classifier was implemented using:

- Laplace smoothing
- Log-probabilities to prevent numerical underflow
- Stratified 80/20 train–test split

Test Set Performance Metrics

- **Accuracy:** 1.00
- **Precision:** 1.00
- **Recall:** 1.00
- **F1 Score:** 1.00

Confusion Matrix

	Predicted Edible	Predicted Poisonous
Actual Edible	All correct	0
Actual Poisonous	0	All correct

Part 2B - Comparison with sklearn

Using the same training and test split, the sklearn model performance

- **Accuracy:** 1.00
- **Precision:** 1.00
- **Recall:** 1.00
- **F1 Score:** 1.00
- **Confusion Matrix:** Identical to the custom implementation.

Part 2C - Comparison of Implementations

Are the predictions identical?

Yes, the predictions are identical.

Why are they identical?

Both models give the same results because they use the same Naive Bayes method, including Laplace smoothing and log-probabilities to avoid numerical issues. The mushroom dataset is easy to separate because the features clearly show whether a mushroom is edible or poisonous. Because the data is clean and clear, both models learn in the same way and make the same predictions.

Bonus Question

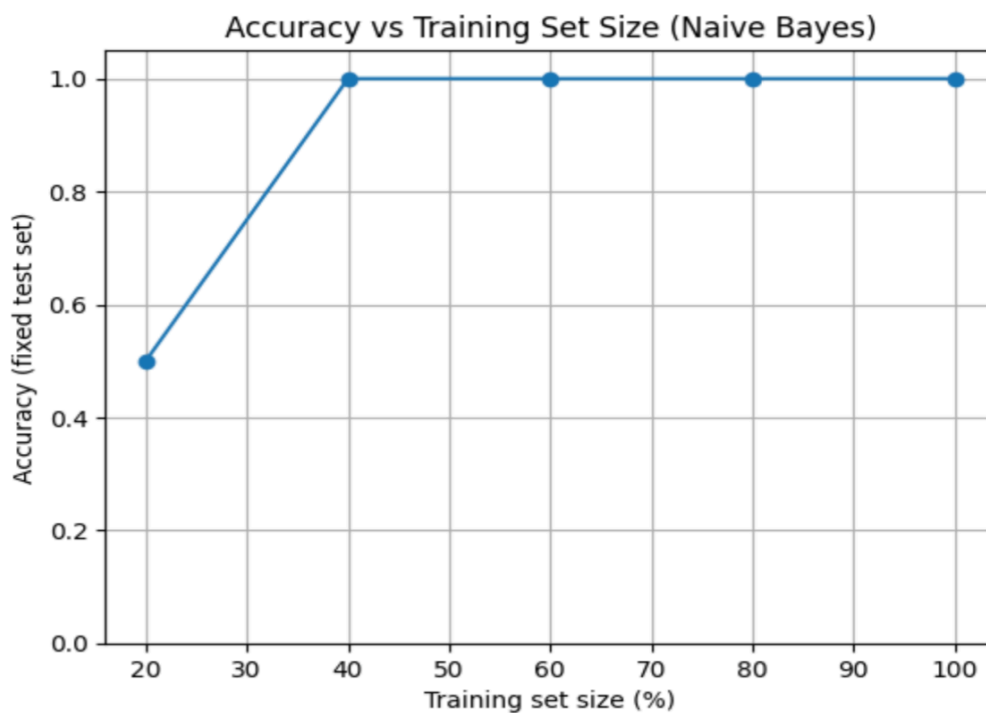
The classifier was trained using varying fractions of the training data:

Training Size Test Accuracy

20%	0.50
40%	1.00
60%	1.00
80%	1.00
100%	1.00

Observations

- With only 20% of the training data, accuracy was 50%.
- From 40% onward, accuracy reached 100%.
- This shows that even a moderate amount of data is sufficient for Naive Bayes to learn the class distributions effectively on this dataset.



The plot confirms that performance stabilizes quickly as training size increases.